American Computer Science League

2025-2026 ● Contest 4: Shorts Problem Solutions ● Senior Division

1. Graph Theory

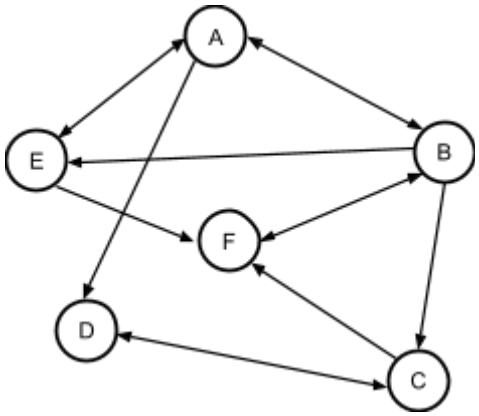
The adjacency matrix raised to the 2nd power for paths of length 2 is as follows:

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1` & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

There are 49 total and 9 are not 0 so there are 40 that are 0.

C. 40

2. Graph Theory



By inspection, there are 8 cycles in this directed graph that contain vertex B: BAB, BADCFB, BAEFB, BCFB, BEAB, BEADCFB, BEFB, BFB. Cycles AEA and CDC do not contain vertex B.

E. 8

3. Digital Electronics

The digital electronics circuit translates to the Boolean algebra expression:

$\overline{(AB + B)\,\overline{\overline{BC}} + C}$

$= \overline{\overline{AB + B}} + \overline{\overline{\overline{BC}}} + C$

$= \overline{A}\overline{B} * \overline{B} + BC + C$

$= (\overline{A} + \overline{B})\,\overline{B} + BC + C$

$= \overline{A} * \overline{B} + \overline{B} + BC + C$

$= \overline{B}(\overline{A} + 1) + C(B + 1)$

$= \overline{B} + C$

This is FALSE for (*,1,0), so 6 triples make it TRUE.

D. 6

4. Digital Electronics

The translated Boolean algebra expression is:

$\overline{((A + B) \oplus B) + \overline{\overline{BC}} + C}$

This simplifies to:

$= \overline{(A + B) * \overline{B} + \overline{\overline{(A + B)} * B} + \overline{\overline{BC}} + C}$

$= \overline{A\overline{B} + B\overline{B} + \overline{A} * \overline{B} * B + \overline{\overline{BC}} + C}$

$= \overline{A\overline{B} + \overline{\overline{BC}}} + C$

$= \overline{A\overline{B}} * \overline{\overline{\overline{BC}}} + C$

$= (\overline{A} + B) * BC + C$

$= \overline{A}BC + BC + C$

$= BC + C$

$= C$

C. C

5. Assembly Language

```
A       DC          3
B       DC          7
        LOAD        A
        MULT        B
        STORE       P
        DIV         =10
        STORE       C
        MULT        =-10
        ADD         P
        STORE       E
        ADD         C
        MULT        =10
        STORE       D
        LOAD        C
        MULT        =100
        ADD         D
        ADD         E
        STORE       ANS
        PRINT       ANS
        END
```

This program makes P = A*B = 21. Then C=2 and E=1. D =10*C+E = 30 so ANS = C*100+D+E = 231. In fact, this program finds the product of any double 2-digit number times any 1-digit number, in this case 33*7.

A. 231

6. Assembly Language

```
G       DC          0
L       DC          0
        READ        P
TOP     READ        N
        LOAD        N
        BE          STOP
        LOAD        P
        SUB         N
        BL          CONT
        LOAD        G
        ADD         =1
        STORE       G
        LOAD        N
        STORE       P
        BU          TOP
CONT    LOAD        L
        ADD         =1
        STORE       L
        LOAD        N
        STORE       P
        BU          TOP
STOP    LOAD        G
        MULT        =10
        ADD         L
        STORE       ANS
        PRINT       ANS
        END
```

This program counts how many are greater than or equal to the previous value (G) and how many are less than the previous value (L). Then it prints G*10+L. G will equal 4 (6, 9, 7, 7) and L will equal 2 (4, 3) so the PRINT statement prints 42. It stops with a value of 0 which is not compared to anything. The 9 is never read into memory.

C. 42